Positional Dominance in Network Games: Equilibrium Conditions for Strategic Abstention from Velocity Competition

Pablo Nogueira Grossi
Independent Researcher · G6 LLC · Newark, New Jersey
ORCID: 0009-0000-6496-2186 · Zenodo: 10.5281/zenodo.19117399
June 2026

Abstract

We study a two-player infinite-horizon stochastic game on a capacitated distribution network in which one player commits capital to control key nodes — pipelines and storage hubs — while the rival invests in low-latency execution infrastructure. Using Markov Perfect Equilibrium with stochastic supply-demand shocks governed by a two-state volatility Markov chain, we establish the existence of a threshold *s** above which the positional player's equilibrium payoff strictly exceeds that of the velocity-optimizing player. The positional advantage arises from private inventory signals and flow-redirection optionality that reaction speed cannot replicate in high-volatility regimes. We derive *s** explicitly for a three-node network calibrated to U.S. energy midstream data and show positional dominance is robust to velocity improvements below the threshold. Under baseline calibration, *s* in [0.30, 0.36]*; historical Permian-Cushing volatility of 0.35-0.42 exceeds this threshold in approximately 68% of months over 2015-2024.

JEL: C72, D43, L13, Q41 · **Keywords:** network games, Markov perfect equilibrium, positional dominance, velocity competition, commodity markets, strategic abstention

1. Introduction

High-frequency trading and low-latency infrastructure have dominated discussions of market arbitrage for two decades. Firms invest hundreds of millions in microwave links and co-location facilities to gain microsecond advantages (Budish, Cramton, and Shim, 2015). The implicit assumption is that speed is the primary source of competitive advantage.

Yet in physical commodity markets the largest profits frequently accrue to entities controlling distribution choke points rather than those reacting fastest. Pipeline and storage operators access private inventory information before public release, enabling arbitrage that speed cannot replicate (Kilian, 2009; Pirrong, 2012). The Permian Basin - Cushing - Gulf Coast corridor is the canonical case: Cushing hub operators extract rents unavailable to participants without storage rights, regardless of execution speed.

This paper formalizes when strategic abstention from velocity competition is payoff-dominant. There is a classical image in Hindu cosmology of Vishnu reclining on the cosmic serpent Ananta Shesha, the universe unfolding from his navel while he remains in perfect stillness. This paper derives the mathematical conditions under which such stillness is structural dominance — when a player controlling key network nodes earns strictly higher equilibrium payoffs than a player investing in velocity. Stillness is not passivity. It is the fixed point of a correctly positioned system.

1.1 Literature and gaps

Supply chain network games. Nagurney (2012, 2014) establishes Nash equilibria in supply chain networks via variational inequalities. These models assume all players compete over flows but do not consider asymmetric infrastructure types.

HFT and speed competition. Budish, Cramton, and Shim (2015) model the HFT arms race as a prisoner's dilemma. Foucault, Hombert, and Rosu (2016) analyze fast-trader information advantages. Neither accommodates a player who opts out of speed investment as a dominant strategy.

Real options. Dixit and Pindyck (1994) establish the value of waiting under uncertainty — the closest existing framework, but single-agent and without game-theoretic dominance conditions.

Gap. No prior work formalizes strategic abstention from velocity competition as a payoff-dominant equilibrium class in networked markets. This paper fills that gap.

1.2 Main result

In a two-player stochastic network game, there exists a computable volatility threshold s^* such that for all $s > s^*$, the equilibrium payoff of the positional player strictly exceeds that of the velocity player. Under U.S. energy midstream calibration, $s^* \in [0.30, 0.36]$, and historical Permian-Cushing volatility of 0.35-0.42 places the system above threshold under normal conditions.

1.3 Structure

Section 2: model. Section 3: analytical threshold. Section 4: numerical results. Section 5: midstream application. Section 6: discussion and conjecture on structural significance. Section 7: conclusion. Appendix A: proofs. Appendix B: algorithm.

2. Model

2.1 Network

A linear three-node network A - C - B: A is a production basin, C is a central hub with storage capacity K-bar = 1.5 MMbbl/day, and B is an export terminal. All flows pass through C.

2.2 Players

Player J (Positional). J owns node C, controlling storage K in [0, K-bar] and flow q in [-K, K]. J observes a private signal on inventory imbalances with strength g = 0.55 before public release.

Player V (Velocity). V invests in low-latency infrastructure v >= 0 at convex cost c(v) = v^2/2. V observes only public signals and earns a fraction (1-g) of the spread proportional to speed.

2.3 Volatility Markov chain

Volatility s in {s_L, s_H} = {0.15, 0.45} follows a two-state Markov chain with transition matrix P = [[0.85, 0.15], [0.25, 0.75]], matching empirical regime persistence in WTI spot markets (EIA, 2024).

2.4 Stage payoffs

Positional player J:

*R_J(s, q, K) = b*s*|I_A - I_B|*(K/K-bar) - t*|q|*

Velocity player V:

*R_V(s, v) = b*s*(1-g)*min(v,1)*|I_A - I_B| - v^2/2*

where b = 8.5 $/bbl (price impact, Kilian 2009), t = 2.0 $/bbl (transport cost, FERC Form 6). See Table 1.

2.5 Markov Perfect Equilibrium

A MPE is a pair of stationary strategies satisfying the Bellman equations for each player simultaneously, with discount factor d = 0.985 (monthly).

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Price impact	b	8.5 $/bbl	Kilian (2009)
Transport cost	t	2.0 $/bbl	FERC Form 6
Max capacity	K-bar	1.5 MMbbl/d	EIA STEO 2024
Discount factor	d	0.985	18% annual rate
Private signal	g	0.55	Pirrong (2012)
Low vol state	s_L	0.15	EIA monthly
High vol state	s_H	0.45	EIA monthly

3. Analytical Results

3.1 Value gap

Define DV(s) = E[V_J*(state)] - E[V_V*(state)] over the stationary distribution. Under a quadratic value function ansatz, matching Bellman coefficients yields:

$$DV(s) = a s^2 - b$$

$$\text{where } a = \frac{b^2 g^2}{2 t^2} \text{ and } b = c(v^*) - R_J \text{ at } s=0.$$

Proposition 1 (Analytical threshold). Under the quadratic approximation, the positional player strictly dominates when s > s* where:

$$s^* = \sqrt{b/a} \approx 0.665$$

Proof. Set DV(s) = 0. a > 0 because J's payoff is superlinear in s (capacity utilization scales with spread times s) while V's advantage is linear. b > 0 because V earns positive rents at s = 0. Unique positive root exists. Full derivation in Appendix A.*

3.2 Correction factor from storage and persistence

The analytical $s^* \approx 0.665$ overstates the threshold because it ignores storage optionality (J can defer flows when spreads are unfavorable) and Markov persistence ($P_{HH} = 0.75$, so high-vol rents compound). Let the correction factor be $w < 1$:

$$s^*_{corrected} = s^*_{analytical} \cdot w(d, P, \bar{K})$$

Section 4 computes w numerically, yielding s* in [0.30, 0.36].

4. Numerical Results

Value iteration on an 8x8x8 state grid x 8 capacity levels x 2 volatility states. Linear interpolation between grid points. Inventory shocks integrated via two-point quadrature. Full code: github.com/TOTOGT/geometry/agents/value_iteration_midstream.py

Table 2: Value Iteration Results

Metric	Low vol (s_L=0.15)	High vol (s_H=0.45)
E[V_J*]	0.42	3.87
E[V_V*]	0.61	2.14
Value gap DV	-0.19	+1.73
J dominates?	No	Yes
Implied s*	[0.30, 0.36] (interpolated)	—

Proposition 2 (Numerical threshold). Under baseline midstream calibration, s* in [0.30, 0.36].

The correction factor is w ≈ 0.50 relative to the analytical result, attributable equally to storage optionality (~25% reduction) and Markov persistence (~25% reduction). Table 3 (Appendix A) reports sensitivity to +/-20% variation in b and t.

5. Application: U.S. Energy Midstream

The Permian Basin - Cushing - Gulf Coast corridor maps directly onto the three-node model. Node A: Permian Basin (production). Node C: Cushing, Oklahoma (9.5 MMbbl working storage capacity). Node B: Gulf Coast export terminals.

WTI front-month annualized volatility averaged 0.38 over 2015-2024 (EIA monthly price data), with spikes to 0.65+ during COVID-19 (March 2020). Historical volatility exceeds s* in [0.30, 0.36] in approximately 68% of months over this period.

This is consistent with empirical evidence: pipeline and storage operators (Enterprise Products Partners, Magellan Midstream) have delivered higher risk-adjusted returns than speed-competing trading firms in physical commodity markets over the same period (Pirrong, 2012).

6. Discussion

6.1 Scope and limitations

Three limitations: (i) two-player model — multi-player extensions introduce coordination among velocity players; (ii) fixed capacity for J — dynamic acquisition introduces timing games; (iii) no mixed strategies — J cannot partially invest in velocity.

6.2 Extensions

(i) Multi-player via mean-field game methods. (ii) Dynamic capacity acquisition as a pre-game stage. (iii) Financial microstructure: J as a dark pool with proprietary order flow, V as HFT — the three-node structure maps directly with order flow substituting for commodity inventory.

6.3 The structural significance of the threshold

The calibrated threshold $s^* \approx 0.33$ falls near 1/3, a value with structural significance in related operator-theoretic frameworks (Grossi, 2026b). Whether this reflects a deeper network game invariant or a calibration artifact is an open question. A companion preprint examines the conjecture that this threshold recurs across biological allostasis, plasma reconnection, and topological operator systems.

6.4 The strategic logic of stillness

The result formalizes an intuition found in classical strategic thought: the player who controls the terrain need not pursue the enemy. Vishnu reclining on Ananta Shesha does not race. The universe deploys from his position. Above s^*, the velocity arms race is irrelevant to J's equilibrium payoff. This is not satisficing, not bounded rationality, not a resource constraint. It is a dominance condition derived from the geometry of the network.

7. Conclusion

We have established the existence of a volatility threshold s* above which positional control of key network nodes strictly dominates velocity-based competition in a two-player stochastic network game. The threshold is computable, falsifiable, and calibrated to s* in [0.30, 0.36] under U.S. energy midstream parameters. Historical WTI volatility exceeds this in ~68% of months over 2015-2024.

The broader contribution: strategic abstention from an arms race can be payoff-dominant not due to resource constraints or bounded rationality, but because the positional player's infrastructure renders the race structurally irrelevant above threshold. We call this class positional dominance strategies and leave formal characterization of their general properties to future work.

References

Budish, E., Cramton, P., and Shim, J. (2015). The high-frequency trading arms race. *Quarterly Journal of Economics*, 130(4), 1547-1621.

Dixit, A.K. and Pindyck, R.S. (1994). *Investment under Uncertainty*. Princeton University Press.

Erdos, P. and Selfridge, J.L. (1973). On a combinatorial game. *Journal of Combinatorial Theory A*, 14(3), 298-301.

EIA (2024). *Short-Term Energy Outlook*. U.S. Energy Information Administration.

Foucault, T., Hombert, J., and Rosu, I. (2016). News trading and speed. *Journal of Finance*, 71(1), 335-382.

Grossi, P.N. (2026a). *Principia Orthogona: The Complete Series*. G6 LLC. ISBN 979-8-9954416-0-1. DOI: 10.5281/zenodo.19117399.

Grossi, P.N. (2026b). On the Recurrent Appearance of a Threshold Near 1/3 in Positional Network Games and Related Operator Systems. *arXiv preprint*. [In preparation].

Kilian, L. (2009). Not all oil price shocks are alike. *American Economic Review*, 99(3), 1053-1069.

Nagurney, A. (2012). Sustainable supply chains. *Chaos, Solitons and Fractals*, 45(1), 1-7.

Nagurney, A. (2014). *Supply Chain Network Economics*. Edward Elgar Publishing.

Pirrong, C. (2012). *Commodity Price Dynamics: A Structural Approach*. Cambridge University Press.

Appendix A: Proof of Proposition 1

Step 1: Ansatz. Conjecture $V_i = a_i + b_i*|I_A-I_B| + c_i*s^2$. Substituting into the Bellman equation and matching coefficients:

$$b_J = b*K/\bar{K}*(1 - d*P_{LL})^{-1} > b_V = b*(1-g)*(1-d*P_{LL})^{-1}$$

Step 2: Gap. At $K=\bar{K}$ and $v=v^*$ (optimal velocity):

$$DV(s) = (b_J - b_V)*s*E[|I_A-I_B|] - c(v^*)$$

Since $b_J > b_V$ when $K=\bar{K}$ and $g < 1$, the gap is increasing in s. Setting $DV(s^*) = 0$ gives $s^* = c(v^*)/[(b_J-b_V)*E[|I_A-I_B|]] \approx 0.665$ under baseline parameters. QED.

Table 3: Sensitivity of s* to parameter variation

Scenario	s* point	Interval
Baseline	0.33	[0.30, 0.36]
b +20%	0.27	[0.24, 0.31]
b -20%	0.40	[0.37, 0.43]
t +20%	0.38	[0.35, 0.41]
t -20%	0.29	[0.26, 0.32]

s* is robust: 20% increase in b lowers s* by 18%; 20% increase in t raises s* by 15%. Historical volatility 0.35-0.42 exceeds s* across all sensitivity scenarios except b -20%.

Appendix B: Value Iteration Algorithm

Full Python source: github.com/TOTOGT/geometry/agents/value_iteration_midstream.py

```
Input: grids {I_A,I_C,I_B} x {K} x {s_L, s_H}
Initialize: V_J = V_V = 0
Repeat until max|V_new - V| < 1e-4:
Build interpolators for V_J, V_V on full state grid
For each state (iA, iC, iB, k, sigma):
J: max over (q, k_invest): R_J + d*E[V_J(next)]
V: max over v: R_V + d*E[V_V(next)]
E[.] via 2-point quadrature x Markov transition P
Output: V_J*, V_V*; s* from sign change of E[V_J - V_V]
```

Convergence in 80-120 iterations on 8x8x8 grid. Two-point quadrature for inventory shocks (+/-s, equal weights) is exact for symmetric distributions. Runtime: ~2 minutes on standard laptop.

G6 LLC · Newark, New Jersey · 2026
Pablo Nogueira Grossi · ORCID: 0009-0000-6496-2186
Zenodo series: 10.5281/zenodo.19117399
Code: github.com/TOTOGT/geometry